Pendulum swing with metal nut

Run r_20261101_052657_69cd33 · Date 2026-11-01 · Verdict: PASS WITH WARNINGS

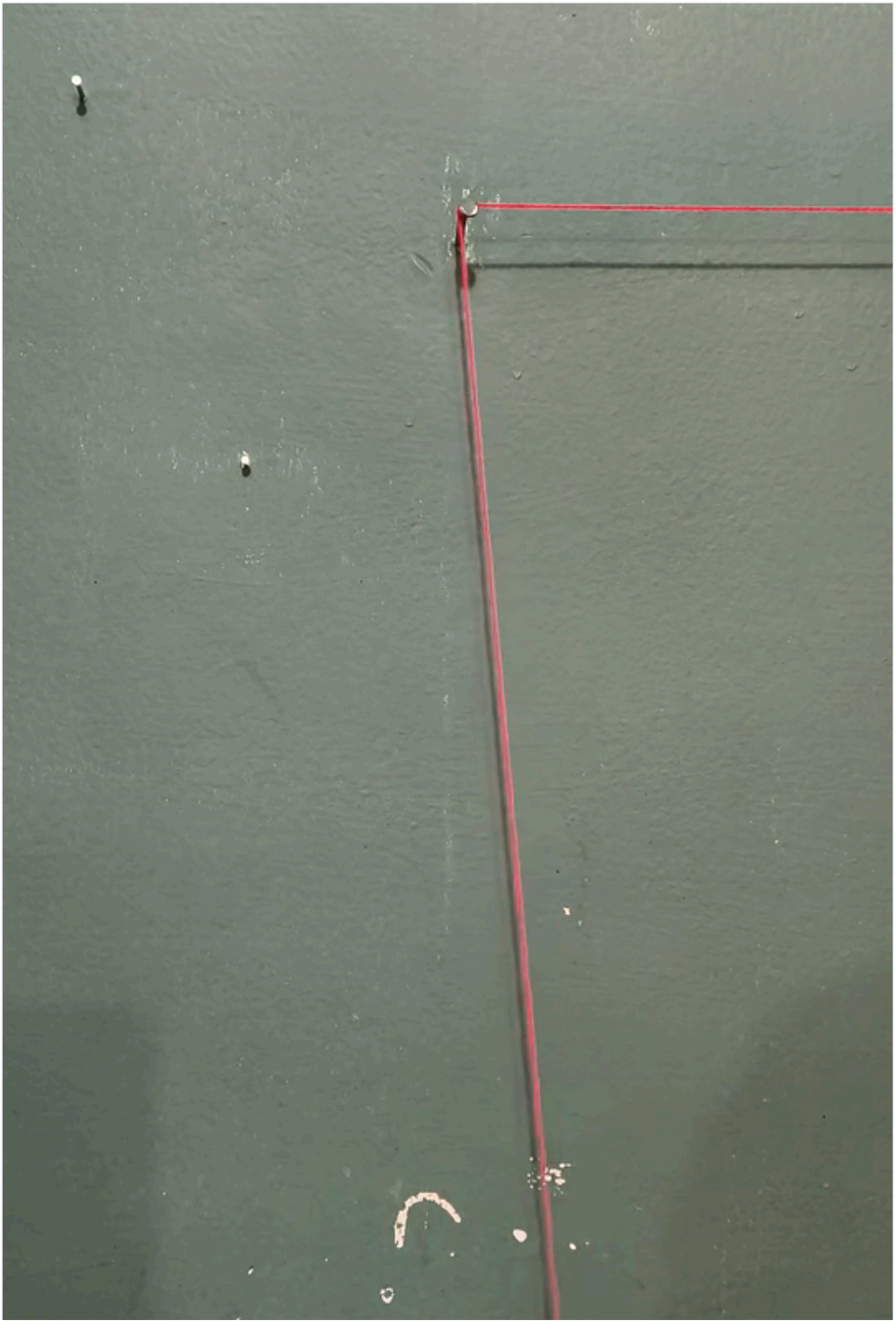
$g = 9.91 \pm 0.10\ m/s^2$

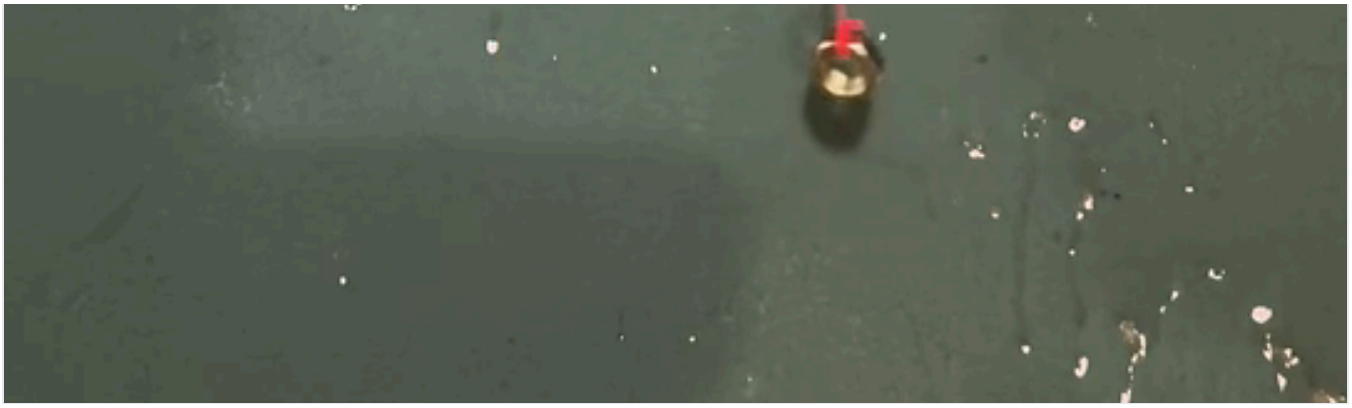
Aim

The aim was to measure gravity using a swinging metal nut and to check the guessed swing time. The experiment uses video tracking of side to side motion.

Setup

The setup was a metal nut hanging from a red string fixed on a green wall. The nut swung side to side while the camera recorded. The string length stated by the student provided the scale for conversion scale = 0.828 mm/px.





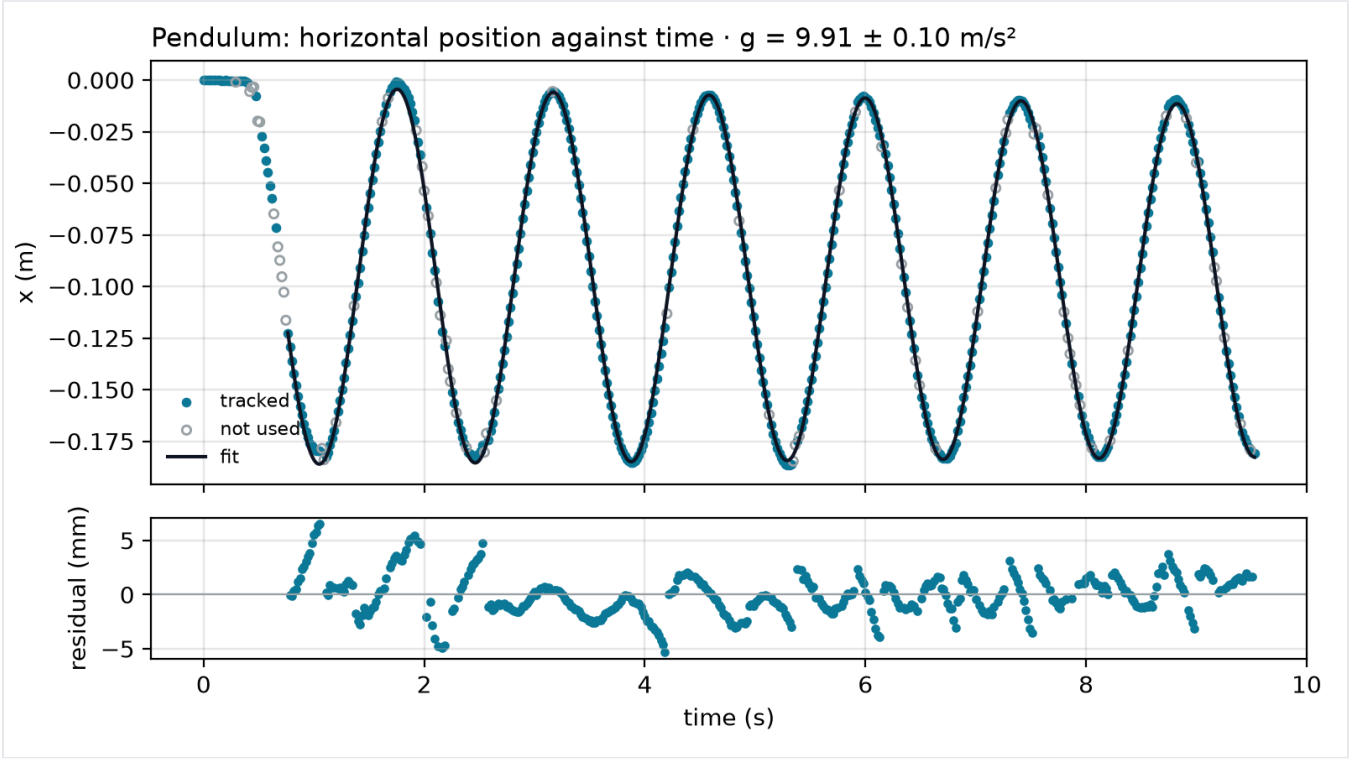
Key frame

Method

The position of the nut was tracked in each video frame across the swing, trimming hand contact at the start. A damped sine curve was fitted to the side to side motion, which follows the trajectory very closely The damped sine fit follows the swing trajectory very closely.. The swing time was estimated with a block bootstrap that respects smooth residual patterns period = 1.4113 ± 0.0014 s, then corrected for swing width period0 = 1.4113 ± 0.0014 s and converted to gravity using the stated string length.

Model	Role in this run
Muse Voice Transcribe	speech → transcript
Muse Spark 1.3	Planner, Analyst, Auditor, Translator, Designer (fallback)
SAM 3.1	object tracking on the video and the ruler
Muse Wonder 1.0	not used in this run
Muse Spark Open (open weights, via Baseten)	Designer

Results



Trajectory

Swing period

The measured oscillation repeats steadily. The measured period is given by the fit period = 1.4141 ± 0.0004 s, with bootstrap support period = 1.4113 ± 0.0014 s and width corrected value period0 = 1.4113 ± 0.0014 s. The close fit supports that the timing is robust.

Gravity value

Using the period and the string length scale, the gravity result is g = 9.91 ± 0.10 m/s². Comparison with the reference shows close agreement a difference of +1.33% from the reference value. The scale basis is documented separately scale = 0.828 mm/px.

Hypothesis evaluation

The guess was about one and a half seconds. The measured period is shorter period = 1.4141 ± 0.0004 s, with corrected support period0 = 1.4113 ± 0.0014 s. The formal test does not support the guess not supported.

Error analysis

Main uncertainties come from the string length, which relies on a spoken value without a visible ruler, and from pivot localisation on the wall. Motion blur caused missing detections in some frames, and a wide swing plus air drag leaves a smooth pattern in residuals, though the bootstrap accounts for this and the fit remains very close The damped sine fit follows the swing trajectory very closely.. Scale agreement could not be checked because only one scale method exists scale = 0.828 mm/px. These effects are included in the reported interval g = 9.91 ± 0.10 m/s².

Conclusion

The pendulum gives a gravity value consistent with the reference within the stated interval g = 9.91 ± 0.10 m/s² a difference of +1.33% from the reference value. The result rests on a clean timing measurement period = 1.4141 ± 0.0004 s and a well fitting model The damped sine fit follows the swing trajectory very closely., limited mainly by length uncertainty.

Next experiment

Pendulum Period vs String Length / Periode Bandul vs Panjang Tali

If string length is increased while keeping mass and small angle the same, the oscillation period will increase from 1.26 s to 1.79 s to 2.19 s to 2.52 s, following period proportional to the square-root of length.

- 0.400 m → 1.26 s
- 0.800 m → 1.79 s
- 1.20 m → 2.19 s
- 1.60 m → 2.52 s

Controlled variables: same bob mass and shape, same small release angle, same support and location, same video timing and lighting method

1. Tie the same household bob such as a metal spoon or plastic toy to a thin string. Do not use glass.
2. Make four lengths measured from pivot to centre of bob, fixed firmly to a door frame below two metres height.
3. Pull the bob to one side only a small angle and release without pushing.
4. Record each swing from the side with a phone in slow-motion with bright room light to avoid motion blur.

5. For each length, time ten full back-and-forth swings and divide by ten to get the period. Repeat three times and average.
6. Keep the same bob, same angle, same camera and same ruler for scale in all trials.

Safety: Use household objects only, support height at most two metres. · No fire, no mains electricity, no glass bob. · Children must have adult supervision.

Appendix A: Claim Ledger

#	Claim	Evidence
c01	The gravitational acceleration from the pendulum is 9.91 ± 0.10 m/s^2 with confidence interval [9.71, 10.11].	der_04
c02	The measured oscillation period is 1.4141 ± 0.0004 s with confidence interval [1.4133, 1.4148].	der_01
c03	The amplitude corrected period is 1.4113 ± 0.0014 s with confidence interval [1.4084, 1.4141].	der_02
c04	The measured gravity differs from the reference by +1.33% with value 9.91 ± 0.10 m/s^2.	cmp_01
c05	The bootstrap estimate for the period is 1.4113 ± 0.0014 s with interval [1.4084, 1.4141].	ci_01
c06	The scale from the spoken string length was used for conversion.	cal_02
c07	The damped sine fit follows the swing trajectory very closely.	fit_01
c08	The hypothesis test outcome is not supported with value 1.4141 ± 0.0004 s.	cmp_02

Appendix B: Gate table

Gate	Status	Diagnosis	Message
G1 Track integrity soft	warn	low_coverage_blur	Track coverage is 88%: the object was not found in some frames, usually from motion blur. Use slow motion or more light.
G2 Scale consistency soft	warn	single_scale	Only one scale method is available, so scale consistency cannot be checked.
G3 Fit quality soft	warn	smooth_model_mismatch	The fit is very close (R^2 0.9992, RMS 2.11 px), but the residuals still follow a smooth pattern (Durbin–Watson 0.27), as a wide swing or air drag causes. The uncertainty is computed with a block bootstrap, which allows for this.
G4 Physical plausibility soft	pass		Passed.
G5 Ledger completeness hard	pass		Passed.

Appendix C: Data sample

t (s)	x (m)	y (m)	valid	flag
0.000000	0.000000	0.000000	1	ok
0.381818	-0.000853	0.001415	1	ok
0.763636	-0.122836	-0.001521	1	ok
1.145455	-0.176805	-0.012265	1	ok
1.527273	-0.048214	0.006991	1	ok
1.909091	-0.019551	-0.002404	1	ok
2.290909	-0.161285	-0.012457	0	outlier
2.672727	-0.149677	-0.006486	1	ok
3.054545	-0.016636	0.003960	1	ok
3.436364	-0.064241	-0.003337	1	ok
3.818182	-0.182047	-0.018319	1	ok

t (s)	x (m)	y (m)	valid	flag
4.200000	-0.113192	0.001299	0	outlier
4.581818	-0.007111	0.003130	1	ok
4.963636	-0.107541	-0.006641	1	ok
5.345455	-0.184715	-0.015240	0	outlier
5.727273	-0.065735	0.004520	1	ok
6.109091	-0.023213	-0.000863	1	ok
6.490909	-0.148015	-0.012427	0	outlier
6.872727	-0.159154	-0.007020	1	ok
7.254545	-0.029687	0.002252	1	ok
7.636364	-0.049145	-0.000134	1	ok
8.018182	-0.173315	-0.018811	1	ok
8.400000	-0.124137	0.000692	1	ok
8.781818	-0.010451	0.002990	1	ok
9.163636	-0.090845	-0.002237	0	outlier

Generated from the run's saved files. Built with Meta SAM 3.1 · Muse Spark · Muse Spark Open · Auditor: muse-spark-1.3